The role of scalar forcing in direct numerical simulations of turbulent atmospheric clouds

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Direct numerical simulation (DNS) is a promising approach to fill knowledge gaps regarding the microscale evolution of turbulent clouds. The governing equations for the flow of ambient air are solved in Eulerian framework, assuming homogeneous and isotropic turbulence. Two scalar fields, namely temperature and vapor mixing ratio, are transported by the turbulent velocity field. Aerosol particles and cloud droplets are tracked in Lagrangian fashion. It is known that forcing is required to sustain turbulence in a DNS model. Although momentum forcing is common in DNS studies of clouds, forcing in the scalar fields has not gained enough attention. The present work implements forcing in the scalar fields while the momentum field is also forced. Mixing between the cloud and environment is simulated. Our results show that forcing drives the mean scalar fields to equilibrium faster and suppresses microscale phase change. Scalar fluctuations are sustained through forcing and probability distributions of the forced scalar fields are non-Gaussian. It is observed that scalar forcing counteracts the homogenization of scalar fields caused by flow turbulence. The intensity of flow turbulence as well as the magnitude of scalar forcing influence the microphysics of cloud droplets and aerosol particles. As the scalar fields are coupled with particles in our model, they have reduced diffusivity and higher effective Schmidt number. Their spectra include a viscous-convective regime which is further widened when forcing is applied. It is shown that a DNS model with scalar forcing can capture extreme fluctuations in the supersaturation and super-adiabatic droplet growth, which are witnessed in natural clouds.

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NOMENCLATURE

α	= imposed scalar variance [K^2 for temperature and $(g/kg)^2$ for vapor mixing ratio]
β	= mean scalar gradient [K/m for temperature and g/kgm for vapor mixing ratio]
c_p, C_d	= specific heat of air [J/(kg.K)], rate of condensation or evaporation [1/s]
δ_{k,k_f}, δ_T, δ_{q_v}	= delta functions [-] in the Fourier space, temperature field, and vapor field, respectively
$\epsilon_{T,in}$, $\epsilon_{q_v,in}$	= dissipation rates in the initial temperature [K^2/s], and vapor [$(g/kg)^2$/s] fields
ϵ, η	= turbulent kinetic energy dissipation rate [m^2/s^3], Kolmogorov scale [m]
η_B, E_T, E_{q_v}	= Bachelor scale [m], temperature spectrum [K^2m], vapor spectrum [$(g/kg)^2$m]
$\boldsymbol{F}_b$, $\boldsymbol{F}_c$	= buoyancy force [N], external force in the momentum field [N]
F_T, F_{q_v}	= external force in the temperature field [N], external force in the vapor field [N]
$\boldsymbol{g}$, $\boldsymbol{k}_f$	= acceleration vector due to gravity [m/s^2], forcing wavenumber vector [1/m]
k_{min}, k_{max}	= minimum wavenumber [1/m], maximum wavenumber [1/m]
κ, k_T	= hygroscopicity of solute [-], thermal conductivity in air [W/(m.K)]
L_h, λ, M_l	= specific latent heat [J/kg], Taylor microscale [m], molar mass of water [kg/mol]
m_a, μ_T, μ_v	= mass of air [kg], molecular diffusivities [m^2/s] of temperature, water vapor in air
ν, p	= kinematic viscosity of air [m^2/s], instantaneous fluid pressure [N/m^2]
q_v, $q_{v,s}$	= instantaneous water vapor mixing ratio [g/kg], saturation vapor mixing ratio [g/kg]
r_d, r	= dry aerosol radius [μm], wet radius of aerosol particle or cloud droplet [μm]
r_c, R_v	= critical radius [μm], specific gas constant for water vapor [J/(kg.K)]
$R_{q_v'q_v'}$	= auto-correlation coefficient for fluctuating water vapor mixing ratio [-]
ρ_a, ρ_l, S_e	= density of air [kg/m^3], density of water [kg/m^3], environmental supersaturation [-]
S_k, S_c	= particle equilibrium supersaturation [-], critical equilibrium supersaturation [-]
Sc, σ_l	= Schmidt number [-], surface tension of water [N/m]
T, τ_p	= instantaneous temperature [K], particle response time [s]
$\boldsymbol{u}$, u_{rms}	= instantaneous fluid velocity vector [m/s], root-mean-square velocity [m/s]
$\boldsymbol{V}$, $\boldsymbol{X}$	= particle velocity vector [m/s], particle position vector [m]
PDF, TKE	= probability density function [-], turbulent kinetic energy [m^2/s^3]

I. INTRODUCTION

Numerical modeling has become indispensable to study turbulence-cloud-aerosol interactions in the atmosphere since experimental measurements are few and difficult to conduct. The most accurate computational approach is direct numerical simulation (DNS) coupled with Lagrangian particle tracking. This method resolves the flow of turbulent air upto the smallest scale without any kind of turbulence modeling and tracks cloud droplets and aerosol particles individually in Lagrangian fashion.[1] Two thermodynamic (scalar) fields, namely temperature and vapor mixing ratio, are transported in Eulerian framework. DNS models rely on a forcing mechanism to maintain turbulent fluctuations.[2] Many DNS studies (Vaillancourt et al.,[1] Kumar et al.,[3] and Yu Li et al.[4]) and our previous DNS model (Gao et al.[5] and Sharfuddin et al.[6]) applied forcing in the momentum field only. While turbulence was sustained in the velocity field, fluctuations in the scalar fields decayed in those models. However, convective motion, radiative effects, and phase change (latent heat release) sustain thermodynamic fluctuations in real clouds.[7] In the present study, we add forcing in the scalar fields to sustain scalar fluctuations at desired levels. The aim is to realistically represent turbulence-cloud interactions in a DNS model.

Without a turbulence production mechanism, turbulence decays in a numerical model and therefore an external forcing is needed. Eswaran and Pope[2] demonstrated that forcing in the Fourier space produces a statistically stationary velocity field. Janin et al.[8] implemented linear forcing in the physical space to maintain homogeneous and isotropic turbulence (HIT). Fourier forcing is applied to a band of low wavenumbers in the Fourier space, but linear forcing is carried out in the physical space with energy injected at all nodal points. Besides momentum forcing, many previous studies have applied forcing in the scalar fields to prevent the decay of scalar fluctuations. Overholt and Pope[9] applied mean gradient forcing in the physical space and showed that the fluctuating scalar field reaches a statistically stationary state. Gotoh and Watanabe[10] implemented mean gradient forcing in a passive scalar field transported by HIT. Daniel et al.[11] proposed a forcing term in the physical space based on chemical reaction analogy and showed that this approach keeps the scalar field within a predefined bound. Chen and Cao,[12] Watanabe and Gotoh,[13] and Kerr[14] carried out scalar forcing in a band of low wavenumbers in the Fourier space. Carroll et al.[15] adopted physical-space linear forcing that imposes a scalar variance and tested it for a range of Schmidt numbers. In the context of atmospheric clouds, Paoli and Shariff[16] applied Fourier forcing in the temperature and vapor mixing ratio fields in addition to the momentum field. Chen et al.[17] forced the scalar fields in a low wavenumber band to study the evolution of mixed-phase clouds.

Scalar forcing plays an important role at the small scales if scalar fields are coupled with particles. Satio et al.[18] examined the relationship between scalar fluctuations and particles using a DNS model with Fourier scalar forcing. The authors reported that the TKE spectrum decays rapidly in the dissipation scale as expected in HIT, but the scalar spectrum remains flat in the intermediate scales. They attributed modification of the scalar spectrum to the presence of particles that are smaller than the Kolmogorov length. Pumir[19] used DNS and mean scalar gradient forcing for a range of Prandtl numbers to investigate the scale dependence of scalar fluctuations. The scalar spectra differ at all scales, including the small scales, and are different in the directions perpendicular and parallel to the mean gradient. However, the spectral profiles follow a slope of -5/3 and no flattening is observed in the inertial subrange due to the absence of particles in their model. The Schmidt number is less than unity for atmospheric flows, but the finite mass of cloud droplets and aerosol particles increases the effective Schmidt number.[20] By applying a DNS model and Fourier forcing in the scalar fields, Watanabe and Gotoh[13] studied the transport of a passive scalar field for which the Schmidt number is unity. The authors reported the existence of a viscous-convective subrange between the inertial and dissipation ranges.

Critical knowledge gap exists regarding the role of scalar forcing in DNS studies of microscale cloud processes such as condensation, activation, evaporation, and deactivation.[6] So, we extend our previous DNS model[6] to include forcing in the temperature and vapor mixing ratio fields. Three scalar forcing methods: Fourier, mean gradient and linear, are implemented. Although scalar forcing had been used along with DNS, those models[9,10,13,15] did not include microscale particles such as cloud droplets and aerosols which are present in our model. We examine the interactions between scalar forcing and microphysics at the large, intermediate, and small scales. Our initial setup consists of two regions separated by a sharp interface unlike the models of Paoli and Shariff[16] and Chen et al.[17] This allows us to simulate turbulent mixing between the cloud and environment. Also, the intensity of flow turbulence is varied and the impacts on scalar fields and microphysics are discussed.

II. GOVERNING EQUATIONS

The equations solved in our model are described in this section. They include the Navier-Stokes equations, the scalar transport equations with forcing methods in the physical and Fourier spaces, and the equations for Lagrangian particle tracking.

A. Turbulent velocity field

The turbulent air in the atmosphere is assumed to be governed by the physical-space Navier-Stokes equations with the Boussinesq approximation:[21]

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c\,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Here, $\boldsymbol{u}$ is the instantaneous fluid velocity vector and p is the instantaneous fluid pressure, $\boldsymbol{F}_b$ is the buoyancy force, and $\boldsymbol{F}_c$ is the external force applied to sustain flow turbulence. The definitions of $\boldsymbol{F}_b$ and $\boldsymbol{F}_c$ can be found in Ref. 6.

B. Scalar transport and scalar forcing

The thermodynamic variables or scalar quantities in this problem are temperature, water vapor mixing ratio, and supersaturation. The transport equations for temperature and water vapor mixing ratio fields are written in the Eulerian framework as:[5]

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_h}{c_p}C_d + \mu_T\nabla^2 T + \delta_T F_T\,, \tag{3}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -C_d + \mu_v\nabla^2 q_v + \delta_{q_v}F_{q_v}\,. \tag{4}$$

where C_d quantifies the time rate of mass exchange between liquid water and water vapor and is described by Eq. (24). The C_d is positive during condensation and negative during evaporation. The terms $L_h C_d/c_p$ and $-C_d$ are latent heat terms in Eqs. (3) and (4), respectively. The F_T and F_{q_v} are the forcing terms for temperature and water vapor mixing ratio, respectively; and the δ_T and δ_{q_v} are the delta functions defined as

$$\delta_T = \begin{cases} 1 & if\ \dfrac{L_h}{c_p}C_d > 0 \\ -1 & if\ \dfrac{L_h}{c_p}C_d < 0 \end{cases}\,. \tag{5}$$

$$\delta_{q_v} = \begin{cases} 1 & if - C_d > 0 \\ -1 & if - C_d < 0 \end{cases}\,. \tag{6}$$

The introduction of δ_T and δ_{q_v} ensures that the corresponding forcing terms act in the direction of the C_d.

We implement three scalar forcing methods. The first is Fourier forcing for which the forcing term is constructed in the Fourier space as:[22]

$$\hat{f}_\theta(\boldsymbol{k},t) = \epsilon_\theta \frac{\hat{\theta}(\boldsymbol{k},t)}{\sum_{k_f \epsilon_k} \left|\hat{\theta}(\boldsymbol{k}_f,t)\right|^2} \delta_{k,k_f} \,. \tag{7}$$

$$\epsilon_\theta = \epsilon_{\theta,in} m_\theta \tag{8}$$

Above, $\hat{\theta}(\boldsymbol{k},t)$ is the Fourier-transformed scalar field and θ is a placeholder for any scalar. The $\epsilon_{\theta,in}$ is the dissipation rate in the initial scalar field and m_θ is a multiplication factor. The forcing wavenumber range is $1 \leq k_f/k_{min} \leq 4.12$, where k_f is the forced wavenumber and $k_{min} = 2\pi/L = 12.27$ is the minimum wavenumber using domain length $L = 0.512\ m$. The second method is mean gradient forcing for which the forcing term is defined in the physical space as:[9]

$$F_{\theta,MG} = w\beta \tag{9}$$

where w is the instantaneous z-velocity and β is the mean scalar gradient. The unit of β is for K/m for temperature forcing and g/kgm for vapor forcing. The β is calculated by matching $F_{\theta,MG}$ with $\hat{f}_\theta^{-1}(\boldsymbol{k},t)$ as:

$$\beta = \langle \frac{\hat{f}_\theta^{-1}(\boldsymbol{k},t)}{w} \rangle \tag{10}$$

Note that $\hat{f}_\theta^{-1}(\boldsymbol{k},t)$ is found by transforming $\hat{f}_\theta(\boldsymbol{k},t)$ back to the physical space. The third method is linear scalar forcing and the forcing term is constructed in the physical space as:[15]

$$F_{\Theta,LN} = \frac{1}{2}\frac{\langle \chi(\Theta) \rangle}{\sigma_\Theta^2}\Theta \tag{11}$$

The rescaled scalar field Θ is defined as[15]

$$\Theta = \theta\sqrt{\frac{\alpha}{\sigma_\theta^2}} \tag{12}$$

where α is the imposed scalar variance. The quantity $\chi(\Theta)$ is the dissipation rate of Θ and is defined as[15]

$$\chi(\Theta) = \frac{\partial \sigma_\Theta^2}{\partial t} \tag{13}$$

We find the α as follows:

$$\hat{f}_\Theta^{-1}(\boldsymbol{k},t) = \frac{1}{2}\frac{\langle \chi(\Theta) \rangle}{\sigma_\Theta^2}\Theta = \frac{1}{2}\frac{\langle \chi(\Theta) \rangle}{\sigma_\Theta^2}\theta\sqrt{\frac{\alpha}{\sigma_\theta^2}}$$

$$\alpha = \left(\frac{2\hat{f}_\Theta^{-1}(\boldsymbol{k},t)}{\langle \chi(\Theta) \rangle}\frac{\sigma_\Theta^2}{\theta}\right)^2 \sigma_\theta^2 \tag{14}$$

Note that the α is updated at every grid point and every timestep, whereas the β is constant in space and updated at every timestep. The β and the α could not be specified a priori in our model unlike in Refs. 9 and 15 because we have two scalar fields with different mean values. Hence, the mean gradient and linear forcing functions are matched with the Fourier forcing function.

We solve the combined (mean plus fluctuating) scalar fields as shown in Eqs. (3) and (4). Since forcing affects both the fluctuating and mean scalar fields, we define scalar forcing functions in a way that avoids unphysical results. For example, a multiplication factor is used in Eq. (8) so that Fourier forcing is neither too weak nor too strong. It is expected that the mean supersaturation decreases and increases to zero (saturation level) during condensation and evaporation, respectively. We find that a strong forcing can cause the mean supersaturation to be less or greater than zero. On the other hand, a weak forcing cannot sustain scalar fluctuations at a desired level. The multiplication factor for temperature (m_T) is larger than that for vapor mixing ratio (m_{q_v}) because the mean temperature is larger than the mean vapor mixing ratio (Table III). Also, the scalar forcing term (F_T or F_{q_v}) interacts with the latent heat term. If the former is larger than the latter and they have opposite signs, condensation might take place instead of evaporation and the vice versa, altering the physics of the problem. The delta functions (Eqs. (5) and (6)) confirm that the F_T or F_{q_v} and the C_d have the same sign. In that case, the δ_T and δ_{q_v} augment the changes in the mean temperature and vapor mixing ratio, respectively.

C. Growth of cloud droplets and aerosol particles

In the atmosphere, inter-conversion of aerosol particles and cloud droplets takes place.[23] When the environment is supersaturated, aerosol particles condense, grow beyond the critical radii, and activate into cloud droplets. On the contrary, cloud droplets evaporate, shrink below the critical radii, and deactivate into aerosol particles in a subsaturated environment. The critical radius criterion distinguishes between aerosol particles and cloud droplets in our model.[6] To account for curvature and solute effects, we apply the kappa-Kohler theory[24] which defines the particle equilibrium supersaturation S_k as

$$S_k = \frac{A}{r} - \frac{\kappa r_d^3}{r^3 - r_d^3(1-\kappa)}. \tag{15}$$

where r_d is the dry aerosol radius, r is the particle (wet) radius, and κ is the hygroscopicity of the solute. Parameter A is defined as:[6]

$$A = \frac{2\sigma_l M_l}{RT\rho_l} . \tag{16}$$

The critical radius (r_c) and critical equilibrium supersaturation (S_c) can be found as:[6]

$$r_c = \sqrt{\frac{3\kappa r_d^3}{A}} , \qquad S_c = \frac{2}{\sqrt{\kappa}}\left(\frac{A}{3r_d}\right)^{\frac{3}{2}} . \tag{17}$$

The motion of each aerosol particle or cloud droplet is represented in the Lagrangian fashion, which consists of the kinematics and the Newton's law. The simplified equations used for the Lagrangian position and velocity can be written as[5]

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t) , \tag{18}$$

$$\frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{\left(\tau_p\right)_i}[\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)] + \boldsymbol{g} . \tag{19}$$

where the subscript 'i' denotes the i-th aerosol particle or cloud droplet, $\boldsymbol{V}$ is its velocity vector, and $\boldsymbol{X}$ is its position vector. $\boldsymbol{u}_i$ is the instantaneous fluid velocity vector at the position of particle i and is obtained through a trilinear interpolation of the Eulerian field. The particle response time τ_p is given by[5]

$$\left(\tau_p\right)_i = \frac{2\rho_l r_i^2}{9\rho_a \nu} . \tag{20}$$

The change in particle (aerosol or cloud) radius during condensation (or evaporation) is described by[26]

$$\frac{dr_i(t)}{dt} = \frac{G}{r_i(t)}\left(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\right)_i . \tag{21}$$

where S_e is the environmental supersaturation and is related to q_v and T by

$$S_e = \frac{q_v}{q_{v,s}} - 1 , \tag{22}$$

where $q_{v,s}$ is the saturation water vapor mixing ratio.[27] The S_e is a Eulerian quantity in Eq. (22) but interpolated to the particle position in Eq. (21). The growth parameter G is expressed as[26]

$$G = \frac{1}{\frac{L_h\rho_l}{k_T' T}\left(\frac{L_h}{R_v T} - 1\right)(1 + S_k) + \frac{\rho_l R_v T}{\mu_v' p_{sat}}} , \tag{23}$$

where k_T' and μ_v' are the modified thermal conductivity and modified water vapor diffusivity,[28] respectively. With the foregoing, the C_d is determined as:[5]

$$C_d(\boldsymbol{x},t) = \frac{4\pi\rho_l G}{\rho_a a^3}\sum_{i=1}^{n} r_i(t)\big(S_e(\boldsymbol{X},t) - S_k(\boldsymbol{X},t)\big)_i\,. \tag{24}$$

where n is the number of particles in a grid cell that has a volume of a^3.

III. NUMERICAL DETAILS

We conduct two studies: Study I and Study II.[6] The first is for condensation and activation of aerosol particles into cloud droplets in one physical region of a cloud. The second is for evaporation and deactivation of cloud droplets into aerosol particles in another physical region.[6] Note that the domain length (0.512 m) in our model is much smaller compared to an actual cloud, which is typical for a DNS study because of computational cost. As a consequence, no natural mechanism (convective motion, Dirichlet boundaries, or radiative effects) is available in our model to prevent a complete decay of turbulence. Therefore, external forcing is required in the momentum and scalar fields to sustain turbulent fluctuations.

A. The numerical model

By varying the range of forced wavenumbers,[6] we generate two turbulence levels: 'low' and 'high', and denote them as 'L' and 'H', respectively. Specifically, we apply forcing in the wavenumber bands: $1 \le k_f/k_{min} \le 4.12$ and $1 \le k_f/k_{min} \le 12.12$, respectively. The number of grid points N is 256 in one coordinate direction and the domain length L is 0.512 m. Note that the maximum wavenumber $k_{max}{\sim}128k_{min}$ in our model. The initial TKE spectrum is identical for both turbulence levels and is specified as[29]

$$E(k) = \frac{16}{\sqrt{\pi/2}}\frac{u_0^2k^4}{k_0^5}\exp\left(-\frac{2k^2}{k_0^2}\right). \tag{25}$$

where $k_0 = 4.12k_{min}$ and $u_0 = 0.16\ m/s$ is the root-mean-square (rms) velocity.

Table I lists the statistics of turbulent velocity field such as the turbulent kinetic energy, $\hbar$, its dissipation rate, ϵ, the rms velocity, u_{rms}, the Kolmogorov length scale η, the Taylor micro-scale, λ, the length scale characterizing large eddies, l_0, the forcing Reynolds number, Re_f, the Taylor-scale Reynolds number, Re_λ, and the Reynolds number based on l_0, Re_{l_0}. The abovementioned quantities have been defined in Ref. 6. The integral length scales (l), calculated as shown in Ref. 6, are 0.14 m and 0.08 m for 'low' and 'high' turbulence levels, respectively. The parameter ranges listed in Table I are representative of cloud turbulence.[30]

Table I. Statistics of the velocity fields at 4 s

Turbulence level	$\hbar$ (m^2/s^2)	ϵ (m^2/s^3)	u_{rms} (m/s)	λ (m)	η (m)	l_0 (m)	Re_f	Re_λ	Re_{l_0}
High	0.0094	0.00494	0.1386	0.0169	0.00091	0.184	401.12	156.11	1704.7
Low	0.0025	0.00055	0.0797	0.0261	0.00157	0.227	192.97	138.74	1207.6

The physical model simulated is that of a cubic domain (box) inside which turbulence, cloud droplets, and aerosol particles interact. The boundary conditions are triply periodic, and the domain volume is $0.512^3\ m^3$. At the start of simulations, the domain is divided equally into cloudy and clear-air regions. Cloud droplets or aerosol particles are placed in the cloudy region initially and subsequent turbulent mixing causes them to spread to all over the domain. A slab-like cloud configuration (Fig. 1) is assumed for the initial field,[6] to enable a sharp transition in the water vapor mixing ratio and temperature at the cloud/clear-air interface.

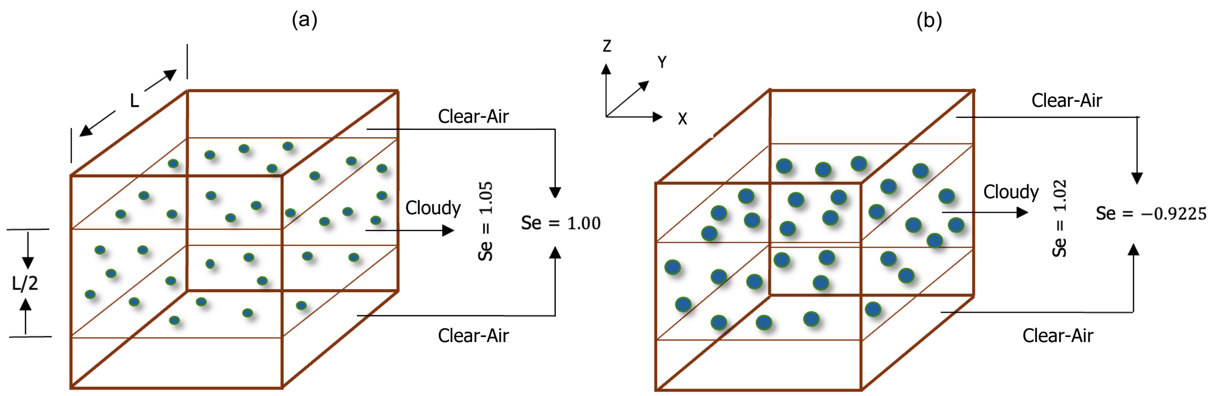


FIG. 1 Initial configurations for (a) Study I, and (b) Study II. The middle half represents the cloudy region while the top and bottom one-fourths constitute the clear-air region. The cloudy region is initially 5% supersaturated in Study I and 2% in Study II. The clear-air region is just saturated in the first and −92.25% subsaturated in the second. The spheres represent aerosol particles in Study I and cloud droplets in Study II.

In Study I, the cloudy region is supersaturated (humid), but the clear-air region is just saturated at initial time. The initial vapor mixing ratio field is specified as:[6]

$$q_v(z,t=0) = \begin{cases} q_v^{max}, & d - L\times 0.5\times 0.5 \le z < d + L\times 0.5\times 0.5 \\ q_{v,e}, & elsewhere \end{cases}. \tag{26}$$

The value of $q_{v,s}$ in Eq. (22) is $3.872\ g/kg$. Above, $q_v^{max} = 1.05\times q_{v,s} = 4.066\ g/kg$, and $q_{v,e} = 3.872\ g/kg$. The length, $d \equiv L/2$ is the width of the cloud slab. To resolve the discontinuous distribution (Eq. (26)), we choose a DNS model based on finite difference[6] instead of a pseudospectral DNS model because the latter could give rise to artificial oscillations (Gibbs phenomenon).[31] The temperature field is initialized as:[6]

$$T(t=0) = \langle T(t=0)\rangle - 0.608\langle T(t=0)\rangle(q_v(z,t=0) - \langle q_v(t=0)\rangle)\,. \tag{27}$$

16×10^6 dry aerosols are randomly placed in the cloudy region at the start of simulations. The number concentration is 119 cm^{-3}. We simulate six cases namely L, L-F, L-MGF, L-LF, H, and H-F, which are summarized in Table II. The identifiers 'L' and 'H' before the dash (-) imply 'low' and 'high' flow turbulence levels, respectively. The identifier 'F' after the dash (-) indicate that scalar forcing is applied in the Fourier space while 'MGF' and 'LF' denote mean scalar gradient forcing and linear forcing, respectively. The β is calculated using Eq. (12) and the α is calculated using Eq. (16). Note that momentum forcing is implemented in the Fourier space for all cases. The size distributions of dry aerosol particles are assumed to be lognormal with geometric mean radius of 0.126 μm and standard deviation of 0.02 μm. The initial volume fraction occupied by dry aerosol particles is approximately 9.98×10^{-11} % and the average Stokes number (S_t) is 7.35×10^{-8}. Dry aerosols (solutes) absorb water from the environment to become aqueous aerosol particles which grow further and activate into cloud droplets in Study I.[6]

TABLE II. Cases in Study I and Study II. Case L has 'low' turbulence level but without scalar forcing. Case L-F has 'low' turbulence level and scalar forcing is implemented in the Fourier space. Case L-MGF applies mean scalar gradient forcing in the physical space and calculates the β adaptively. Case L-LF implements linear scalar forcing in physical space and calculates the α adaptively. Cases H and H-F are analogous to Cases L and L-F except that the turbulence level is 'high'.

Case	Dry aerosol size (μm)	TKE dissipation rate (m^2/s^3)	Turbulence level	Scalar forcing type	Initial particle size in Study I (μm)	Initial particle size in Study II (μm)
L	0.126 ± 0.02	0.0025	Low	N/A	0.126 ± 0.02	10
L-F	0.126 ± 0.02	0.0025	Low	Fourier	0.126 ± 0.02	10
L-MGF	0.126 ± 0.02	0.0025	Low	Mean gradient	0.126 ± 0.02	10
L-LF	0.126 ± 0.02	0.0025	Low	Linear	0.126 ± 0.02	10
H	0.126 ± 0.02	0.0094	High	N/A	0.126 ± 0.02	10
H-F	0.126 ± 0.02	0.0094	High	Fourier	0.126 ± 0.02	10

In Study II, the cloudy region is supersaturated (humid) but the clear-air region is subsaturated (dry) at the start of simulations. The same configuration for vapor mixing ratio as in Eq. (26) is used except that $q_v^{max} = 1.02 \times q_{v,s} = 3.949\ g/kg$ and $q_{v,e} = 0.30\ g/kg$. The temperature field is initialized as described in Eq. (27). A total of 16×10^6 cloud droplets, having a monodisperse size (radius) distribution of 10 μm, are randomly placed in the cloudy region at the start of simulations. Also, dry aerosols are assumed to be present in the core of cloud droplets. The initial volume fraction occupied by cloud droplets is 4.99×10^{-5} % and the average S_t is 4.63×10^{-4}. Cloud droplets evaporate, shrink in size, and deactivate into aerosol particles in Study II.[6] Note that the r_a remains constant at each particle index i but the r changes according to the growth equation (Eq. (21)).

Particles are inserted in the domain after 2 s, when the TKE dissipation rates become statistically stationary. The initial dissipation rates in the temperature ($\epsilon_{T,in}$) and vapor mixing ratio ($\epsilon_{q_v,in}$) fields are included in Table III. The Schmidt number ($Sc = \nu/\mu_T$) is 0.68 for both temperature and vapor mixing ratio. The Bachelor length scales ($\eta_B = \eta/\sqrt{Sc}$) are $0.0019\ m$ and $0.0011\ m$ for 'low' and 'high' turbulence intensities, respectively. However, the effective Sc is higher in our model because scalar fields are coupled with particles via latent heat terms. To balance the grid resolution requirement and computational cost, we carry out all the simulations using 256^3 grid points. The $k_{max}\eta$ values for 'low' and 'high' turbulence levels are 2.46 and 1.43, respectively. The FronTier software package[32] is used to solve the governing equations. The details regarding discretization and numerical techniques can be found in Ref. 6. We have used the Perlmutter supercomputer managed by National Energy Research Scientific Computing Center (NERSC), United States Department of Energy, USA.

B. List of parameters

We list the constants and parameters of our DNS model in Table III. The hygroscopicity parameter (κ) of 0.61 is used to represent ammonium sulfate aerosols.[32]

TABLE III. Parameters with initial values, and other constants

Symbol	Value	Unit	Symbol	Value	Unit
q_v^{max}	4.066 (Study I) 3.949 (Study II)	g/kg	$q_{v,e}$	3.872 (Study I) 0.30 (Study II)	g/kg
L_h	2.5×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_a	1.0	kg/m^3	ν	1.5×10^{-5}	m^2/s
ρ_l	1000	kg/m^3	p	82844.14	N/m^2
$\langle T(t=0)\rangle$	270.75	K	R_v	461.5	J/kg/K
k_T	0.0238	W/m/K	σ_l	0.072	N/m
$\mu_T = \mu_v$	2.16×10^{-5}	m^2/s	M_l	0.018	kg/mol
R	8.314	J/mol/K	S_t	7.35×10^{-8} (Study I) 4.63×10^{-4} (Study II)	-
κ	0.61	-			
$\epsilon_{T,in}$	2.16×10^{-5} (Study I) 7.7×10^{-3} (Study II)	K^2/s	$\epsilon_{q_v,in}$	7.9×10^{-4} (Study I) 2.8×10^{-1} (Study II)	(g/kg)^2/s

m_T	10 (Study I) 1 (Study II)	-	m_{q_v}	0.1 (Study I) 0.1 (Study II)	-

C. Model validation

We ran our model using the parameters of Overholt and Pope[9] who applied mean scalar gradient forcing. The time evolution of normalized scalar variance is plotted in Fig. 2(a). The Cases OP-1996 and OP-128 in Fig. 2(a) represent Overholt and Pope[9] (Run 128.3) and our run, respectively. The number of grid points is 128^3 for both cases. The variance of the scalar field σ_θ^2 is:

$$\sigma_\theta^2 = \frac{\sum_i (\theta_i - \bar{\theta})^2}{N}. \tag{28}$$

where N is the number of grid points. The variance is zero initially for both cases and no latent heat term is present in the scalar transport equation. The scalar variance is normalized by $(\beta L_\epsilon)^2$, where L_ϵ is a turbulent length scale.[9] The fluctuating scalar field is normalized by its standard deviation σ_θ, and the normalized time is t/T_E, where t is the physical simulation time and T_E is the eddy turnover time.[9] We find that the normalized scalar variances for Cases OP-1996 and OP-128 (Fig. 2(a)) are almost equal once they become statistically stationary ($t/T_E \sim 6$). The differences between them from 2 to 6 (abscissa) can be attributed to the transient behavior in respective models.

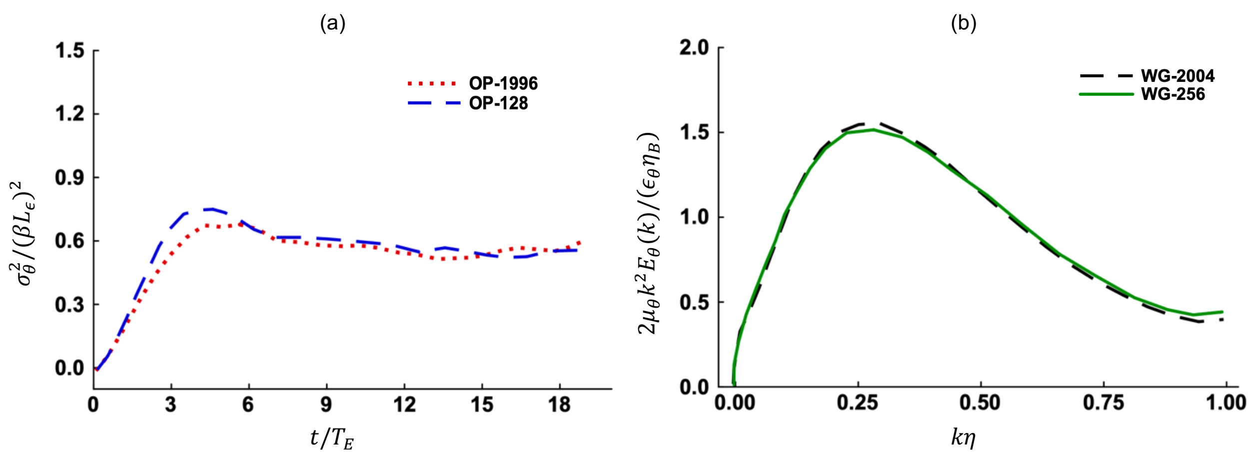


FIG. 2 Comparison of (a) normalized scalar variances and (b) normalized scalar dissipation spectra.

We also validate our model against that of Watanabe and Gotoh[13] who carried out scalar forcing in the Fourier space. According to Ref. 13, the normalized scalar dissipation spectrum is: $2\mu_\theta k^2 E_\theta(k)/(\epsilon_\theta \eta_B)$, where μ_θ, E_θ, and ϵ_θ are diffusivity, spectrum, and dissipation rate in the scalar field θ, respectively. Figure 2(b) shows the normalized scalar dissipation spectra for Cases WG-2004 and WG-256 which represent Watanabe and Gotoh[13] (Run 1) and our run, respectively. The number of grid points is 512^3 for the former but 256^3 for the latter. We find excellent agreement between the two profiles in Fig. 2(b).

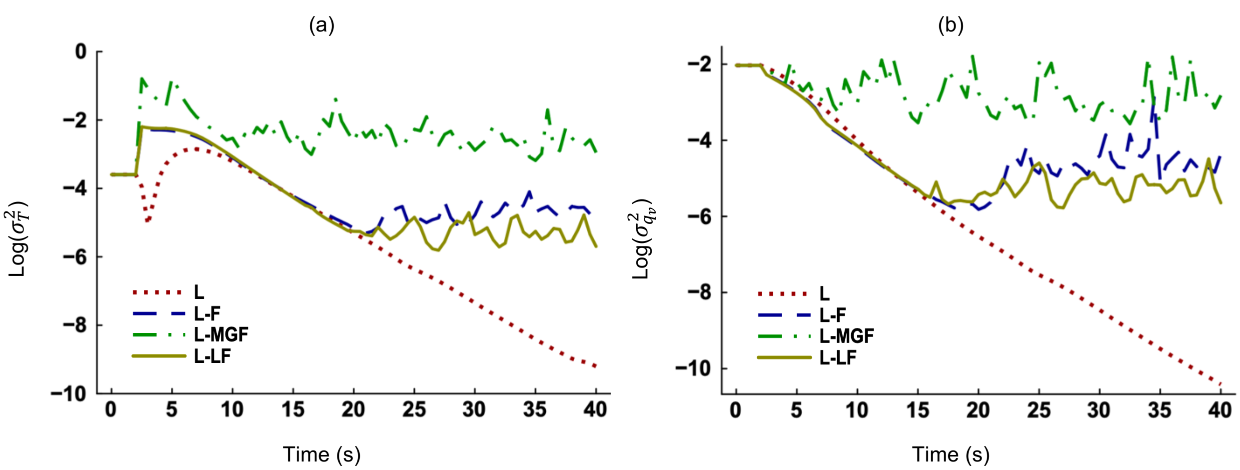
IV. RESULTS AND DISCUSSIONS

The results from our simulations are presented in this section. We discuss the impact of scalar forcing on the fluctuating and mean scalar fields during condensation and evaporation. Then we evaluate the role of scalar forcing on the activation and deactivation processes. The effects of different flow turbulence intensities on scalar fields and microphysics are also discussed, followed by the concluding remarks.

A. The impact of forcing on fluctuating and mean scalar fields

1. Study I (Condensation)

Study I is concerned with condensational growth and aerosol activation. The temporal changes of the variances of fluctuating temperature (σ_T^2), vapor mixing ratio ($\sigma_{q_v}^2$), and environmental supersaturation ($\sigma_{S_e}^2$) are shown in Figs. 3(a)-(c), respectively. We see that the magnitudes of scalar fluctuations are nonzero initially. The forced scalar fields attain statistical stationarity but the unforced scalar fields in Case L continue to decay. The magnitudes of fluctuations are largest for Case L-MGF but very close for Cases L-F and L-LF. The mean gradient forcing term, unlike the linear forcing term, is not matched with the Fourier forcing term at every nodal point. Rather, the mean gradient β is found by spatial averaging (Eq. (10)). In absence of forcing, the fluctuating scalar fields decay due to mixing between the cloudy and clear-air regions. Whereas scalar forcing acts as a source of fluctuations and counteracts the decay.



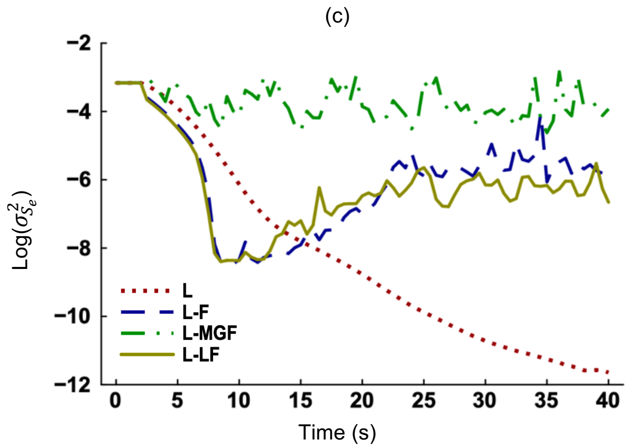


FIG. 3 Time evolution of the variances of (a) fluctuating temperature, (b) fluctuating vapor mixing ratio, and (c) fluctuating environmental supersaturation, in Study I. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and calculates the β adaptively. Case L-LF has linear scalar forcing and calculates the α adaptively. The turbulence level is 'low' for all cases.

The spectra[33] of temperature and vapor mixing ratio are plotted in Figs. 4(a)-(b). They show the distribution of scalar fluctuations across scales. According to the Kolmogorov's theory, small-scale structures should be universal and independent of the large-scale anisotropy.[34] However, small-scale universality is not observed in the scalar fields when forcing is applied. We see that the spectra for Cases L and L-F differ at the intermediate and small scales while those for Cases L and L-MGF differ at all scales. A flat region ($3.0 \leq \log(k) \leq 3.2$) can be identified in the spectra for Case L, which suggests the existence of a viscous-convective regime, typically observed for scalars with Schmidt number (Sc) greater than unity.[35] The Sc for both scalars are less than unity (0.68) in our model without consideration of particles. But the coupling of scalar fields with tiny cloud droplets and aerosol particles correspond to $Sc > 1$ as transport of mass (particles) is slower than transport of momentum. The flat region expands further ($1.8 \leq \log(k) \leq 3.2$) for Cases L-F and L-LF, and the viscous convective regime appears to suppress the inertial convective regime for these two forced cases. Consequently, the spectral slopes deviate from the -5/3 law in the inertial convective regime as seen in Figs. 4(a)-(b). A key feature of higher Sc is the increase of high-wavenumber contribution to the scalar spectrum.[36] We find that fluctuations increase at the intermediate and small scales for Cases L-F and L-LF compared to the unforced case (L). When mean gradient forcing is applied (Case L-MGF), a constant (mean) scalar variance is injected at all physical-space nodal points and fluctuations increase at all scales compared to Case L.

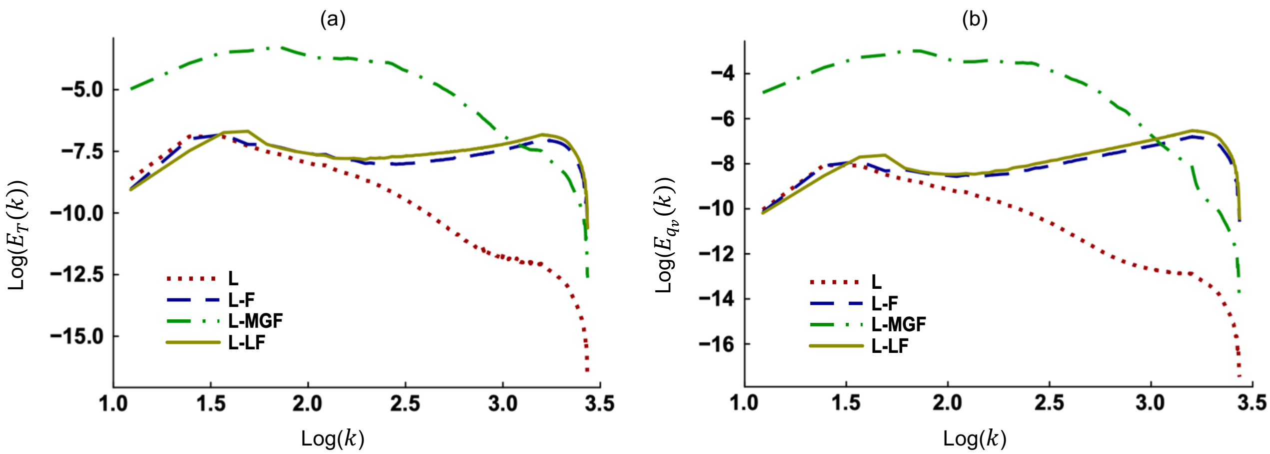


FIG. 4 Spectra of the (a) temperature and (b) vapor mixing ratio fields in a log-log format. The results represent Study I at 25 s.

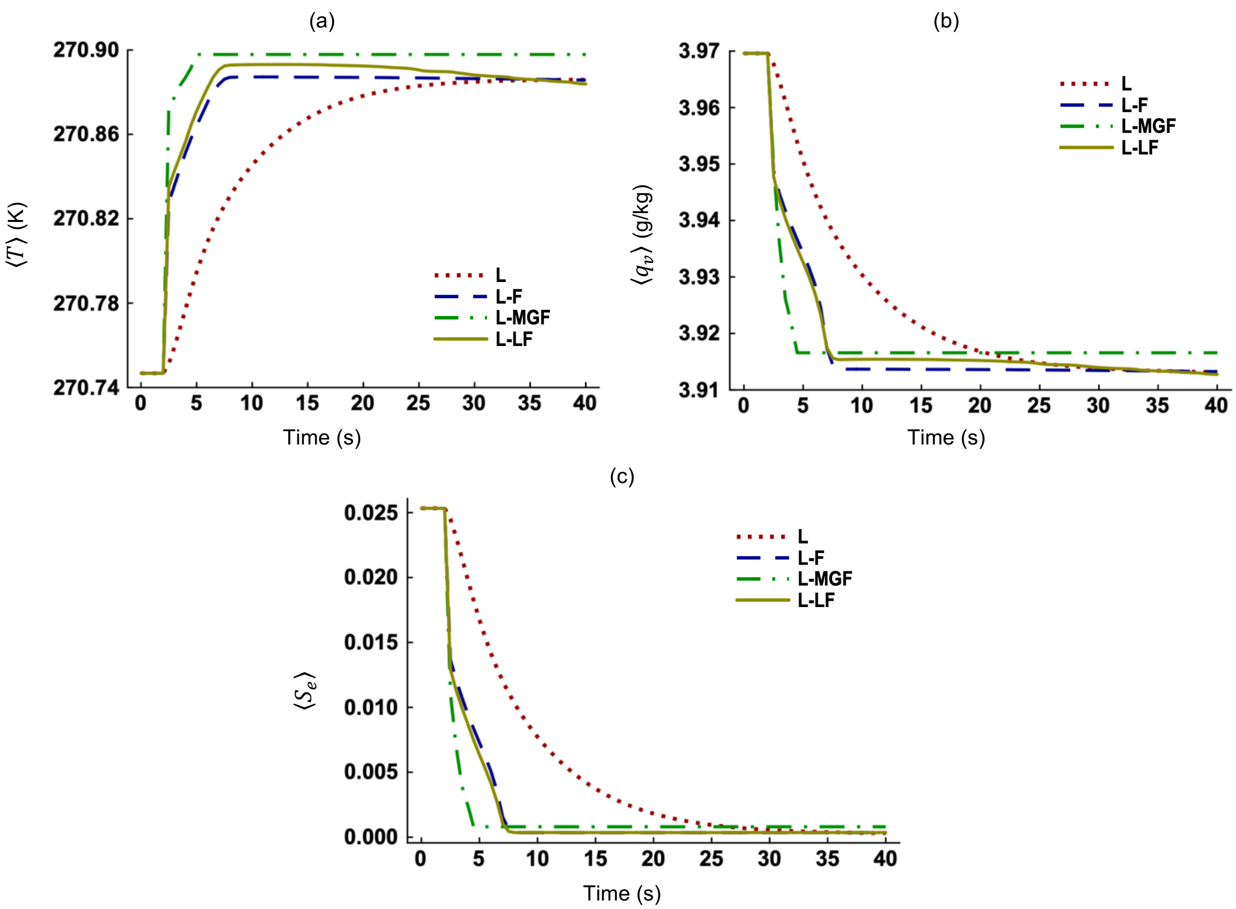


FIG. 5 Time evolution of (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), in Study I. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is 'low' for all cases.

Figures 5(a)-(c) show the temporal evolution of mean scalar fields in Study I. The time rates of increase in the $\langle T \rangle$ and decrease in the $\langle q_v \rangle$ and $\langle S_e \rangle$ are slowest when scalar forcing is absent (Case L). Conversion of water vapor into liquid water during condensation decreases the $\langle q_v \rangle$ and thus the $\langle S_e \rangle$. Figure 5(c) shows that the system reaches equilibrium when the $\langle S_e \rangle$ approaches zero (saturation). Equilibrium is reached by approximately 35 s for Case L and 7 s for the forced cases. We find that forcing causes the mean scalar fields to relax to their equilibrium states faster. This is because phase change drives the evolution of mean scalar fields in our model (Fig. 6(a)) and scalar forcing acts in the direction of phase change.

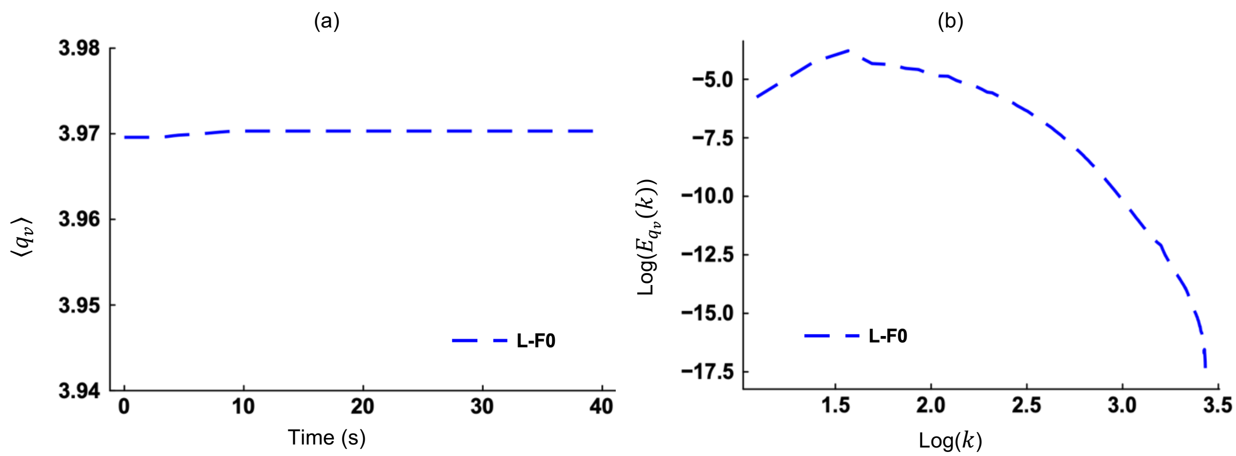


FIG. 6 (a) Time evolution of the mean vapor mixing ratio ($\langle q_v \rangle$) and (b) spectrum of the vapor mixing ratio ($E_{q_v}(k)$) at 10 s in log-log format. The results are for Case L-F0 in Study I. Case L-F0 implements Fourier scalar forcing but excludes particles.

To recognize the effects of particles on scalar fields, we simulate Case L-F0 by setting particle size (r) to zero. When the r is zero, the C_d is zero (Eq. (24)) and the latent heat terms in Eqs. (3) and (4) are zero. We see in Fig. 6(a) that the temporal variation in the $\langle q_v \rangle$ is negligible for Case L-F0. Exclusion of particles prevents mutual conversion of water vapor and liquid water. So, the $\langle q_v \rangle$ remains constant as phase change cannot take place. Figure 6(b) plots the $E_{q_v}(k)$ at 10 s. For Case L-F0, there is a peak in the $E_{q_v}(k)$ which is followed by an exponential decay at the small scales. Thus, the $E_{q_v}(k)$ resembles the spectra observed in HIT.[2] This suggests that Fourier forcing does not cause deviation from homogeneity and isotropy in the scalar fields if particles are absent. But presence of inertial particles modifies the scalar spectra as shown for Cases L-F, L-MGF, and L-LF (Figs. 4(a)-(b)).

2. *Study II (Evaporation)*

Study II focuses on evaporation and deactivation of cloud droplets. The temporal evolution of the relative dispersions of the scalar fields ($\sigma_\theta / \langle \theta \rangle$) in Study II is plotted in Figs. 7(a)-(c). While the scalar fields decay in absence of forcing, the forced scalar fields become statistically stationary. In actual clouds, the magnitudes of fluctuations in the temperature and vapor mixing ratio are much smaller than the mean,[37] which is also the case in our study. The

$\sigma_T/\langle T\rangle$ and $\sigma_{q_v}/\langle q_v\rangle$ for Case L in Study II at 25 s are $1.6e-5$ and $2.97e-3$, respectively. The corresponding ratios for Case L-F are $8.5e-5$ and $2.73e-2$, respectively. So, fluctuations in the temperature and vapor mixing ratio increase with forcing but the standard deviations remain smaller than the means. The ratio $\sigma_{S_e}/\langle S_e\rangle$ is $8.78e-3$ for Case L but 9.04 for Case L-F in Study II at 25 s. Hence, the σ_{S_e} is larger than the $\langle S_e\rangle$) for Case L-F while the opposite is true for the unforced case (L). This finding is consistent with the model of Paoli et al.[16] that applied scalar forcing in the Fourier space. Because the $\langle S_e\rangle$ is much smaller than the $\langle T\rangle$ and $\langle q_v\rangle$, fluctuations in the supersaturation are more pronounced with forcing. Note that the σ_{S_e} often exceed the $\langle S_e\rangle$ in atmospheric clouds.[38]

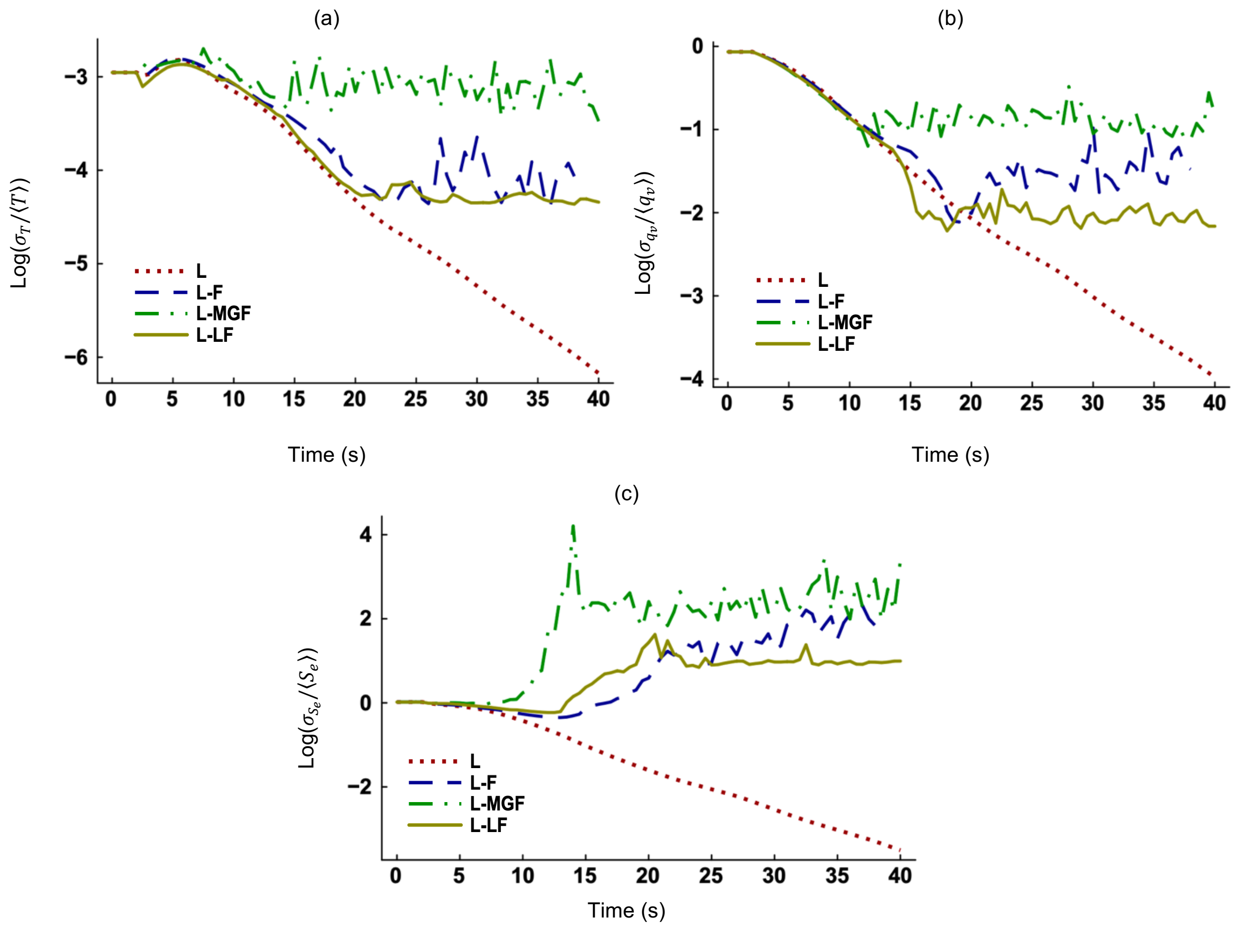


FIG. 7 Time evolution of the relative dispersions of (a) temperature, (b) vapor mixing ratio, and (c) environmental supersaturation, in Study II. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean gradient forcing and Case L-LF has linear forcing. The turbulence level is 'low' for all cases.

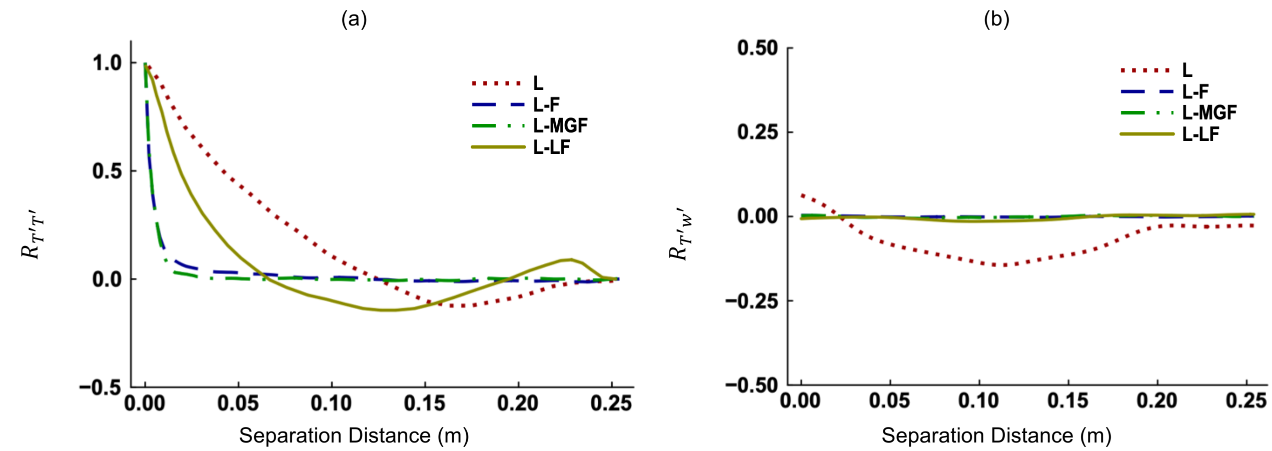


FIG. 8 The correlation coefficients (a) $R_{T'T'}$ and (b) $R_{T'w'}$ in Study II at 25 s.

Spatial correlation coefficient quantifies the mutual dependence between two quantities separated in space. The autocorrelation coefficient $R_{T'T'}$ is defined as[33]

$$R_{T'T'}(\boldsymbol{r}) = \frac{\langle T'(\boldsymbol{x},t)T'(\boldsymbol{x}+\boldsymbol{r},t)\rangle}{\sqrt{\langle\left(T'(\boldsymbol{x},t)\right)^2\rangle\langle\left(T'(\boldsymbol{x},t)\right)^2\rangle}}\,, \qquad \boldsymbol{r} = \Delta x\boldsymbol{i} + \Delta y\boldsymbol{j} + \Delta z\boldsymbol{k}\,. \tag{29}$$

where $\boldsymbol{r}$ is the separation distance vector and T' is the fluctuating temperature. The cross-correlation coefficient $R_{T'w'}$ can be defined similarly with w' being the fluctuating z-velocity. The $R_{T'T'}$ and $R_{T'w'}$ are plotted in Figs. 8(a)-(b), respectively. As separation distance increases, the $R_{T'T'}$ decays more quickly for Cases L-F, L-MGF, and L-LF compared to the unforced case (L). Rapid decorrelation suggests that turbulent structures cannot maintain coherence over a long distance. This happens because fluctuations in the scalar fields are driven by locally injected scalar variances. We see in Fig. 8(b) that the $R_{T'w'}$ is essentially zero and thus the T' and w' are not correlated at all for the forced cases. This confirms that fluctuations in the forced scalar fields are not driven by flow turbulence.

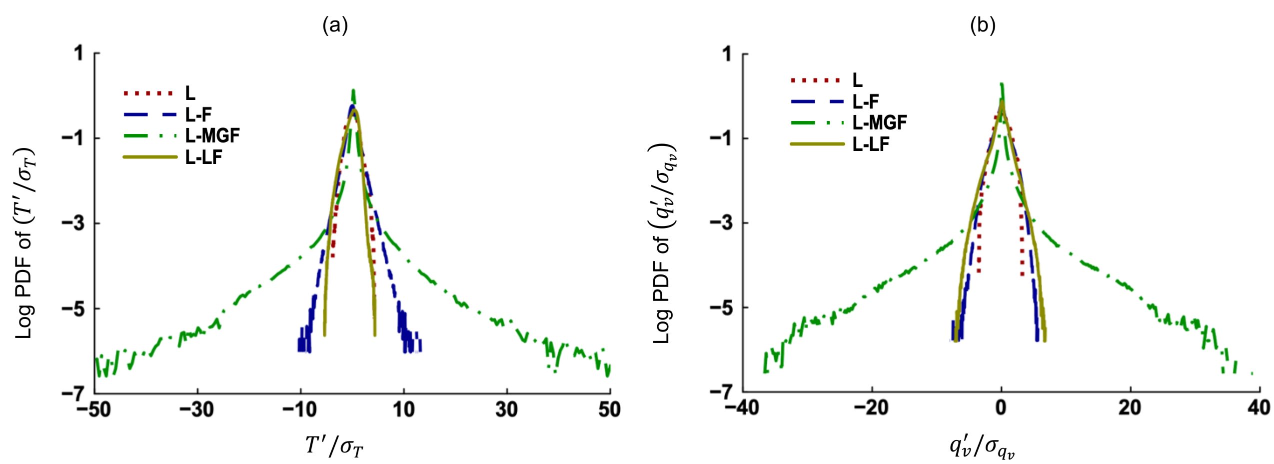
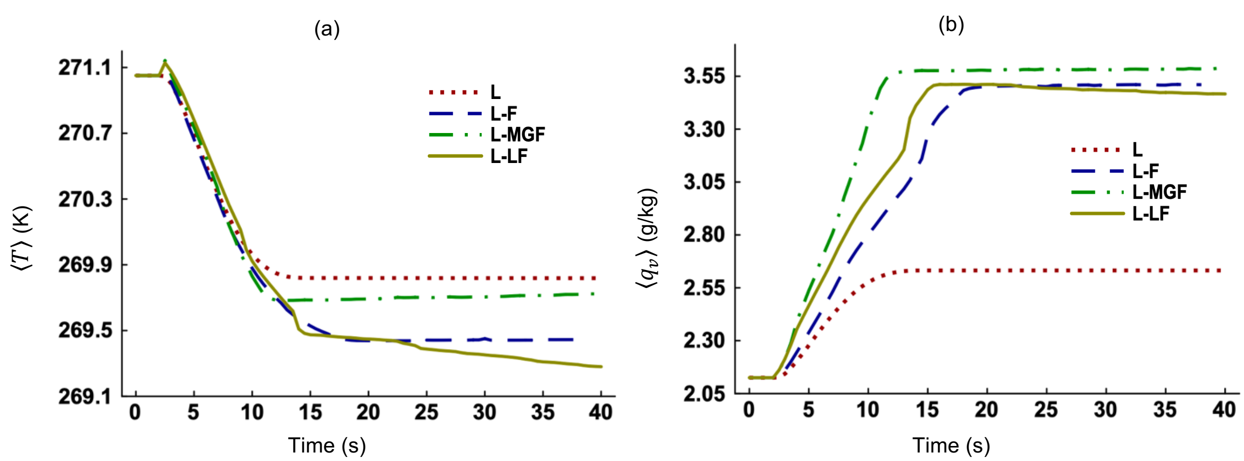


FIG. 9 Log PDFs of the (a) fluctuating temperature (T') normalized by its standard deviation (σ_T), and (b) fluctuating vapor mixing ratio (q_v') normalized by its standard deviation (σ_{q_v}), at 25 s in Study II.

Figure 9(a) shows the PDFs of T'/σ_T, which is the fluctuating temperature normalized by standard deviation, at 25 s for the four cases. Figure 9(b) is analogous to Fig. 9(a) but for vapor mixing ratio. It appears that the normalized PDFs in both figures are broader for Case L-MGF compared to the other cases. To assess how much Gaussian or non-Gaussian the distributions are, we calculate their skewness (γ) and kurtosis (ω) as[34]

$$\gamma = \frac{\langle(\varphi')^3\rangle}{\langle(\varphi')^2\rangle^{3/2}}\,, \quad \omega = \frac{\langle(\varphi')^4\rangle}{\langle(\varphi')^2\rangle^{4/2}}\,, \qquad \varphi = \frac{\theta'}{\sigma_\theta}. \tag{30}$$

The γ of temperature PDFs in Fig. 9(a) are 0.055, 0.28, -0.12, and -0.76 for Cases L, L-F, L-MGF, and L-LF, respectively, while the corresponding ω are 3.21, 6.68, 425.2, and 3.73. The γ of vapor PDFs in Fig. 9(b) are 0.09, 0.003, -0.035, and -0.31, respectively, while the corresponding ω are 3.03, 4.43, 176.2, and 5.84. Note that skewness is 0 and kurtosis is 3 for a perfectly Gaussian distribution. The kurtosis for Case L in both figures are close to 3 but are larger than 3 for the forced cases. So, forcing makes the PDFs of fluctuating scalar fields non-Gaussian. Previous studies of cloud turbulence found that thermodynamic fluctuations can be both Gaussian and non-Gaussian.[39,40] With mean gradient forcing (Case L-MGF), the PDFs in Figs. 9(a)-(b) have heavy tails and their kurtosis are very high. Hence, extreme values are observed for this case although most values are closer to the mean. Physically, extreme fluctuations in the temperature, vapor mixing ratio, and supersaturation occur in natural clouds during updrafts and downdrafts.[41]



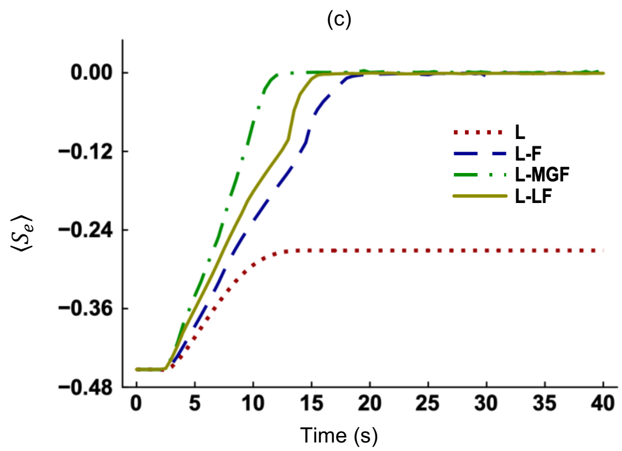


FIG. 10 Time evolution of (a) mean temperature ($\langle T \rangle$), (b) mean vapor mixing ratio ($\langle q_v \rangle$), and (c) mean environmental supersaturation ($\langle S_e \rangle$), in Study II. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is 'low' for all cases.

The temporal evolution of the mean scalar fields for Study II is shown in Figs. 10(a)-(c). We see that the growth and decay profiles are opposite to those in Figs. 6(a)-(c). During evaporation, liquid water converts into water vapor, increasing the $\langle q_v \rangle$ and $\langle S_e \rangle$. The decrease in $\langle T \rangle$ is a manifestation of evaporative cooling. Equilibrium is reached before saturation ($\langle S_e \rangle \sim 0$) for Case L (Fig. 10(c)). Absence of scalar forcing (Case L) leads to faster evaporation and the profile of $\langle S_e \rangle$ settles to an asymptote when all droplets evaporate fully. On the other hand, equilibrium corresponds to saturation for the forced cases and is reached before evaporation is complete.

B. The impact of scalar forcing on cloud microphysics

1. Activation of aerosol particles

To recognize the impact of scalar forcing on condensation (activation), we plot the fraction of aerosol particles that activate into cloud droplets with time in Fig. 11(a). The activation process is very fast between 2 s to 5 s for Case L and slows down afterwards. For the forced cases, the activation fraction is lower because equilibrium is reached before aerosol particles could grow sufficiently to be activated. Interestingly, the activation profile for Case L-MGF decreases slightly after its peak at around 4 s. Thus, a small number of particles deactivate after activation for this case due to negative fluctuation. Figure 11(b) show the time evolutions of the mean radius. The profiles tend to asymptote after exponential increase at early times. For the unforced case (L), particle growth (Fig. 11(b)) and activation fraction (Fig. 11(a)) are highest. Scalar forcing drives the system towards equilibrium faster (Figs. 5 and 10). That leaves less time available for completion of phase change.[42] Hence, particle growth (condensation) is reduced, and activation is suppressed.

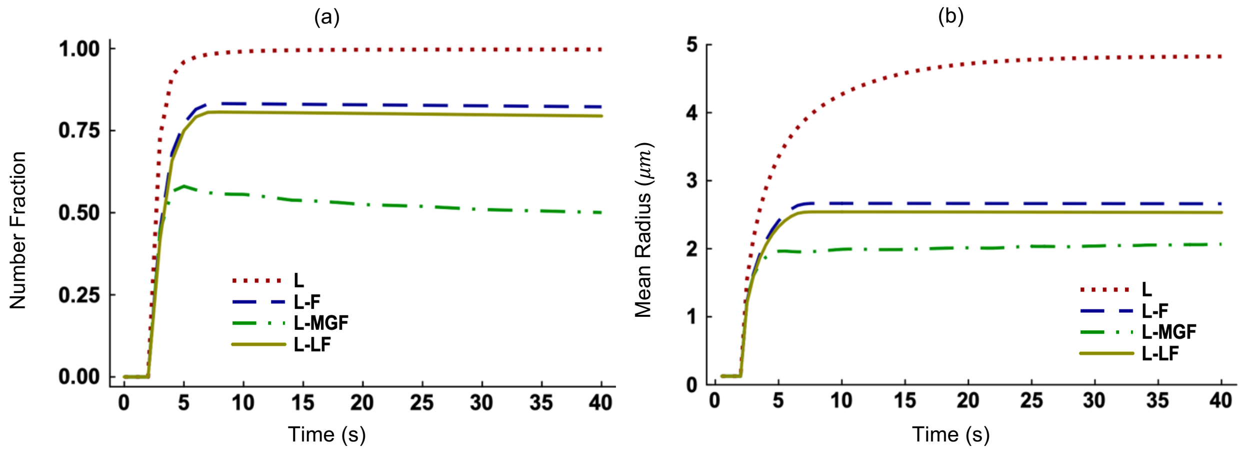


FIG. 11 Time evolution of the (a) number fraction of activated aerosol particles, and (b) mean radius of all particles, in Study I. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is 'low' for all cases.

Fluctuations in the S_e have impact on activation and deactivation. Based on Eq. (21), particle growth is positive if $S_e > S_k$ and negative when $S_e < S_k$. Positive growth means condensation and potential activation while negative growth means evaporation and potential deactivation. Figures 12(a)-(b) show the PDFs of S_e and S_k for Case L-F at 6 s and 25 s, respectively. We find that the $\langle S_e \rangle$ is larger than the $\langle S_k \rangle$ and the PDFs of S_e and S_k overlap at 6 s (Fig. 12(a)). Hence, growth is always positive and activation is mean-dominated[43] for particles that experience $\langle S_e \rangle > \langle S_k \rangle$. But fluctuation-influenced[43] activation or deactivation occurs in the overlapping region. Particles, that have radii closer to the critical radii, will activate or deactivate if fluctuations are positive or negative, respectively. At 25 s, the $\langle S_e \rangle$ is equal to the $\langle S_k \rangle$ while the PDFs overlap (Fig. 12(b)). Since there is no difference between the mean supersaturation values, activation or deactivation is dominated by fluctuations in the S_e and S_k.[43]

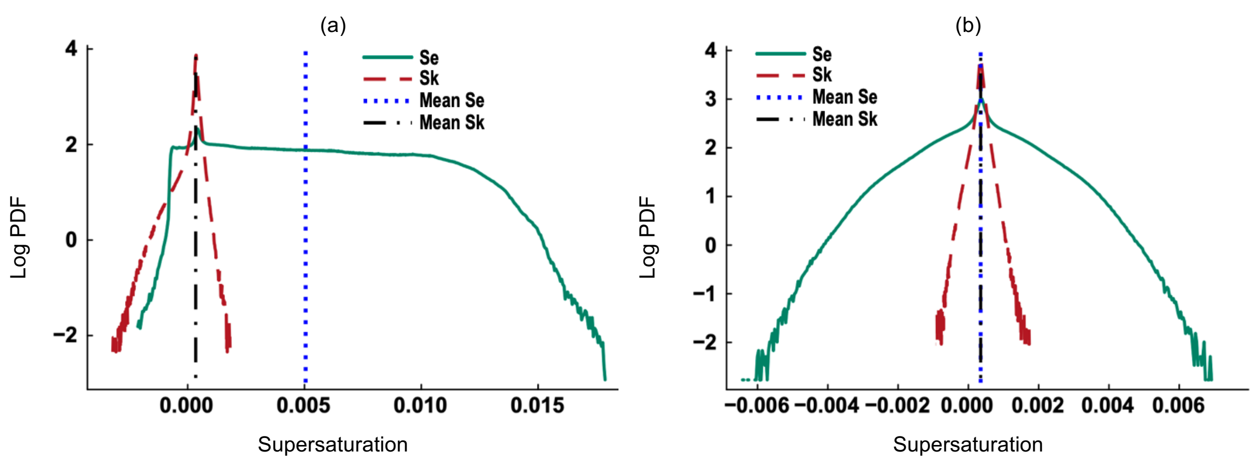
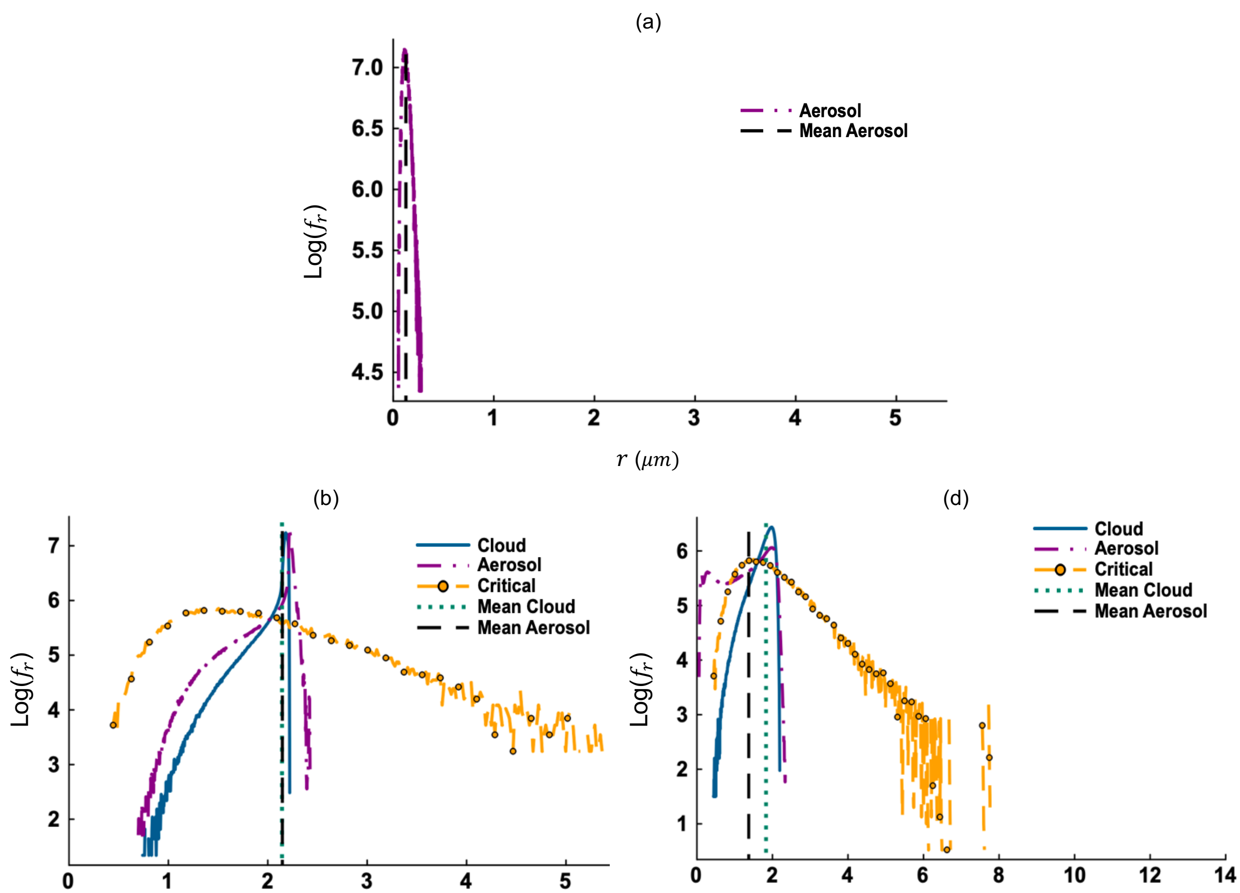


FIG. 12 Log PDFs of instantaneous supersaturation for Case L-F at (a) 6 s and (b) 25 s in Study I. Case L-F implements scalar forcing in the Fourier space. The turbulence level is 'low' for both cases. The vertical lines represent mean values in respective PDFs.

Cloud and aerosol microphysics differ for the unforced and forced cases. Figures 13(a)-(c) show the PDFs of particle radii (f_r) for Case L during condensation (Study I) at 0 s, 3 s, and 10 s, respectively. Figures 13(d)-(e) are analogous to Figs. 13(b)-(c) but represent Case L-MGF. Lognormally distributed dry aerosol particles are present at 0 s (Fig. 13(a)) for both cases. By 3 s, some of the aerosol particles activate into cloud droplets and the f_r of aerosol particles broaden for Case L (Fig. 13(b)). Between 3 s to 15 s, particles grow further, and the mean radii of aerosol particles and cloud droplets increase (Fig. 13(c)). The increase of particle radius and broadening of f_r are also evident for Case L-MGF (Figs. 13(d)-(e)). The f_r of cloud droplets and aerosol particles for Case L are skewed to the left. Thus, some cloud droplets and aerosol particles are much smaller than their mean values. Interestingly, the f_r of cloud droplets are skewed to the right at 15 s for Case L-MGF (Fig. 13(e)). The maximum radius is much larger for this case (13.5 μm) than that for Case L (5.1 μm). Therefore, some cloud droplets experience super-adiabatic growth[44] when mean gradient forcing is applied. However, extreme growth of some droplets is accompanied by reduced growth of other droplets. For example, the mean and minimum radii of cloud droplets for Case L-MGF are smaller than those for Case L at 15 s.



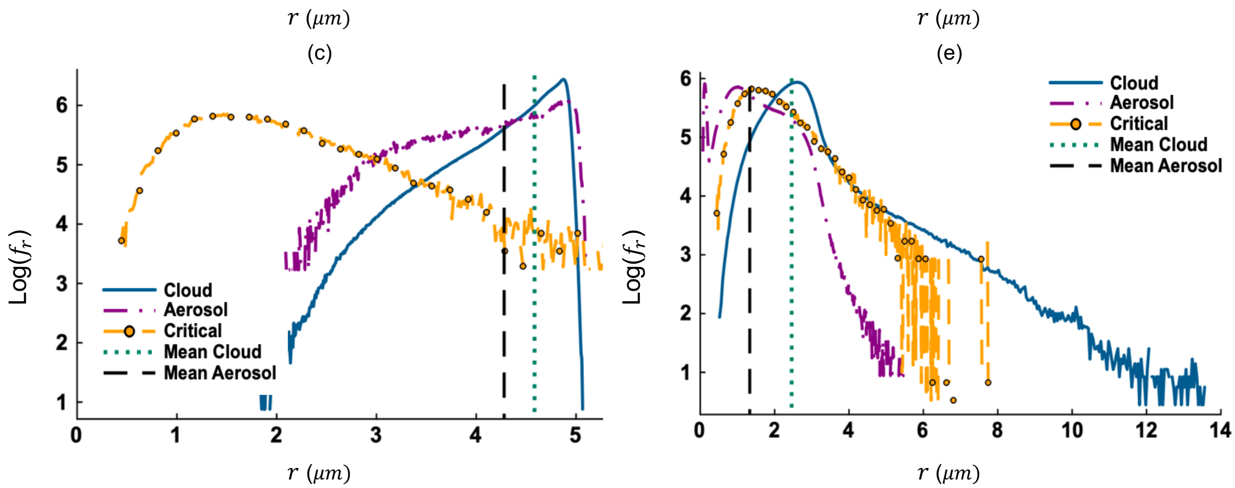


FIG. 13 (a) Log PDF of dry aerosol radii at 0 s. Log PDFs of particle size at (b) 3 s and (c) 15 s for Case L in Study I. Log PDFs of particle size at (d) 3 s and (e) 15 s for Case L-MGF in Study I. The curve with 'circle' mark shows the critical radius (r_c) distribution. Scalar forcing is absent in Case L. Case L-MGF applies mean gradient forcing and calculates the β adaptively.

2. *Deactivation of cloud droplets*

The fraction of cloud droplets that deactivate into aerosol particles with time is shown in Fig. 14(a). We see that deactivation fraction is maximum for the unforced case (L). Figure 14(b) shows the time evolution of mean radius during evaporation. The mean radius decreases most for Case L and least for Case L-MGF. Scalar forcing slows down phase change and evaporative decay in Study II. The profiles in Fig. 14(b) and Fig. 14(a) are apparently in opposite order. As droplet radii decrease by evaporation, the deactivation fraction goes up. Figures 15(a)-(b) are analogous to Figs. 12(a)-(b) but based on the data from Study II. For particles experiencing $\langle S_e \rangle < \langle S_k \rangle$ at 10 s (Fig. 15(a)), growth is always negative, and deactivation is mean-dominated. Also, the PDFs of S_e and S_k overlap (Fig. 15(a)), leading to fluctuation-influenced activation or deactivation in the overlapping region. The $\langle S_e \rangle$ is very close to the $\langle S_k \rangle$ at 25 s (Fig. 15(b)) and the two PDFs overlap. As a result, activation or deactivation is fluctuation-dominated at 25 s.

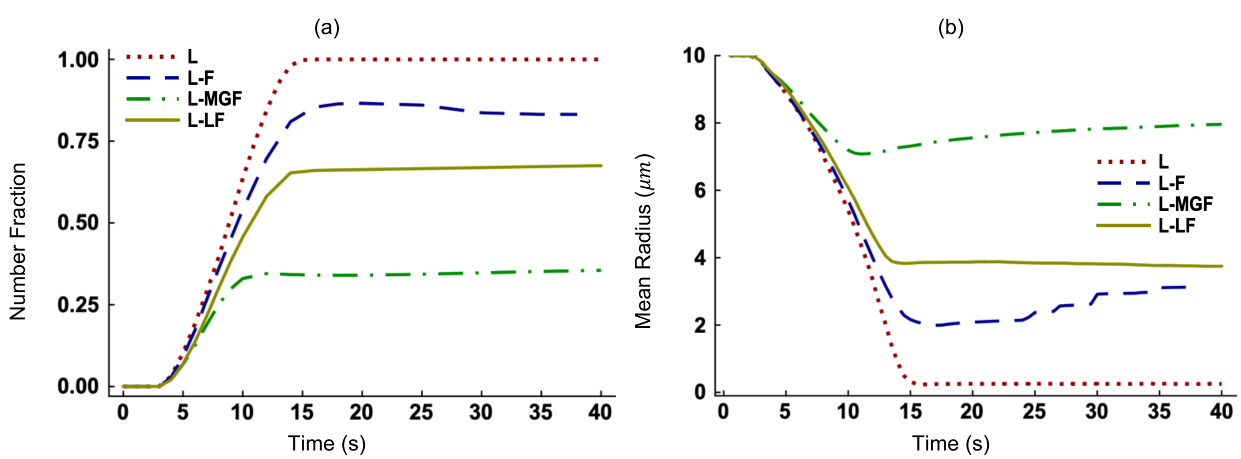


FIG. 14 Time evolution of the (a) number of deactivated cloud droplets, and (b) mean radius of all particles, in Study II. Scalar forcing is absent in Case L. Case L-F implements scalar forcing in the Fourier space. Case L-MGF applies mean scalar gradient forcing and Case L-LF has linear scalar forcing. The turbulence level is 'low' for all cases.

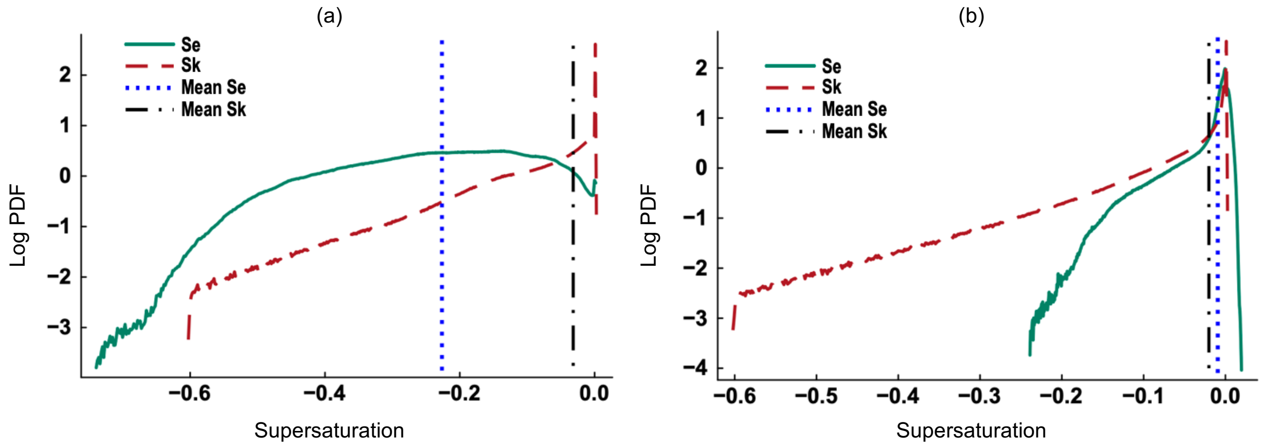


FIG. 15 Log PDFs of instantaneous supersaturation for Case L-F at (a) 10 s and (b) 25 s in Study II. Case L-F implements scalar forcing in the Fourier space. The turbulence level is 'low' for both cases.

The PDFs of particle radii (f_r) are shown in Figs. 16(a)-(e) at three selected times in Study II. As dry subsaturated air mixes with the cloud volume, the mean radii of initially monodisperse ($10\ \mu m$) cloud droplets (Fig. 16(a)) decrease between 0 s to 5 s for Cases L (Figs. 16(b)) and L-LF (Fig. 16(d)). The f_r of cloud droplets broadens and is skewed to the left for both cases at 5 s. On the other hand, the f_r of aerosol particles is skewed to the right. Between 5 s to 15 s, the mean droplet radii decrease for both cases, and the f_r of cloud droplets narrows for Case L (Fig. 16(c)) but not for Case L-LF (Fig. 16(e)). Because of strong evaporation, smaller cloud droplets evaporate completely while larger ones shrink in size when scalar fields are not forced (Case L). By 15 s, the mean droplet radii decrease less for Case L-LF ($\sim 6\ \mu m$) compared to that for Case L ($\sim 3\ \mu m$) since forcing slows down evaporation. Bimodality in the f_r of aerosol particles is evident for both cases, with one peak for smaller aerosol particles and another peak for larger particles. The bimodal size distribution of aerosol particles[45] and the coexistence of aerosols and cloud droplets[46] are observed in atmospheric clouds.

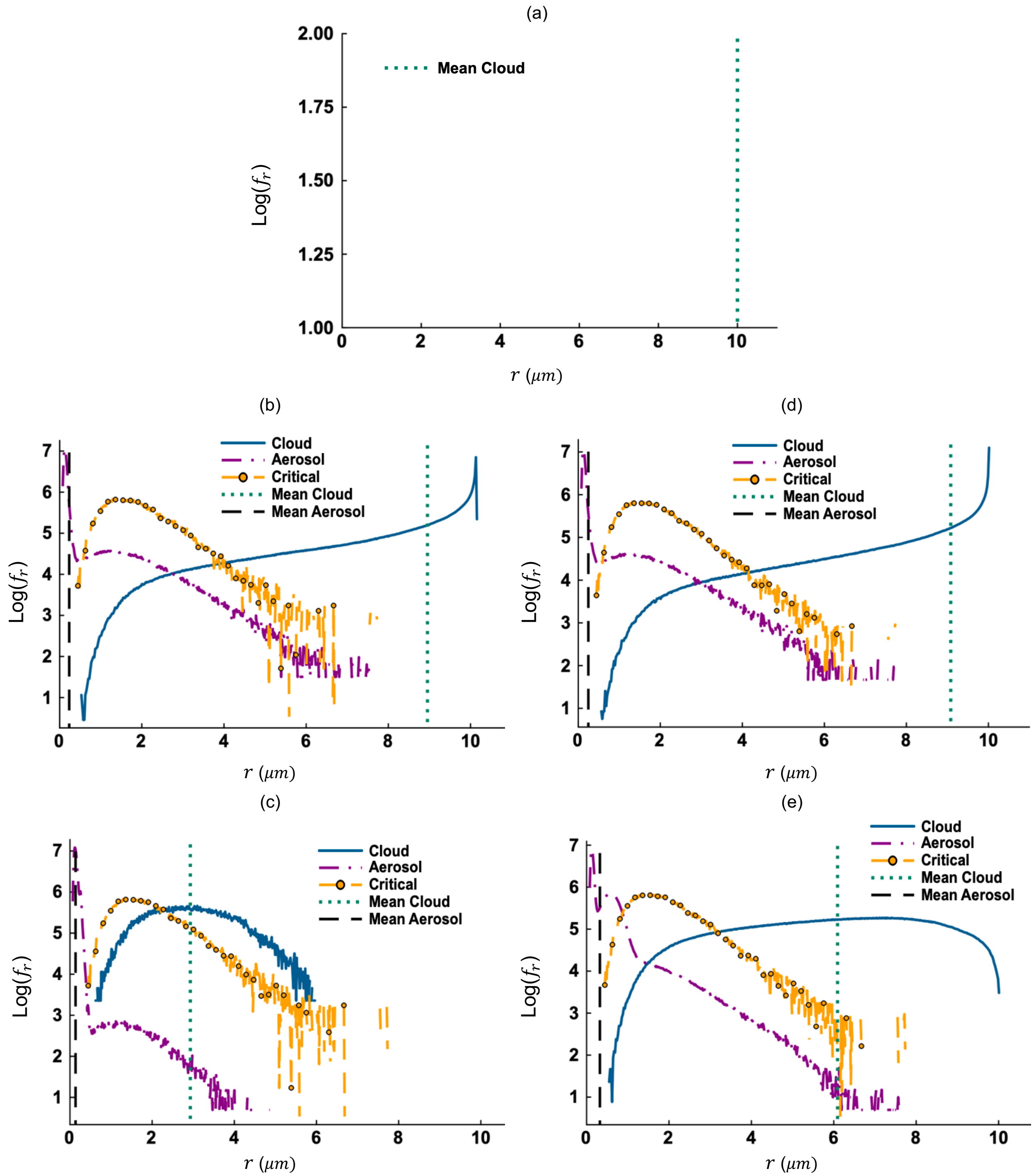


FIG. 16 (a) Log PDF of cloud droplet radii at 0 s. Log PDFs of particle size at (b) 5 s and (c) 15 s for Case L in Study II. Log PDFs of particle size at (d) 5 s and (e) 15 s for Case L-LF in Study II. The curve with 'circle' mark shows the critical radius (r_c) distribution. Scalar forcing is absent in Case L. Case L-LF applies linear scalar forcing and calculates the α adaptively.

C. Variation of flow turbulence intensity in the present model

We simulate Cases H and H-F, where 'H' denotes high flow turbulence level and 'F' denotes scalar forcing in the Fourier space. Figure 17(a) compares the variances of vapor mixing ratio $\sigma_{q_v}^2$ for Cases L, H, L-F, and H-F. The $\sigma_{q_v}^2$ decays faster for Case H compared to Case L. In absence of scalar forcing, higher velocity turbulence homogenizes the scalar fields[6] and reduces fluctuations in the vapor mixing ratio and temperature. When scalar forcing is applied (Cases L-F and H-F), it counteracts turbulent homogenization and the $\sigma_{q_v}^2$ becomes statistically stationary (Fig. 17(a)). The spectra of fluctuating vapor mixing ratio are plotted in Figure 17(b). They appear to converge at the small scales for Cases L and H. But fluctuations increase significantly at the small scales for the forced cases (L-F and H-F). The PDFs of fluctuating z-velocity ($f_{w'}$) and fluctuating environmental supersaturation ($f_{S_e'}$) are shown in Figs. 18(a)-(b), respectively. The $f_{w'}$ for Cases H and H-F are found to be broader than those for Cases L and L-F (Fig. 18(a)). Such broadening is caused by high flow turbulence. The $f_{w'}$ are identical for Cases L and L-F, and for Cases H and H-F because scalar fields and particles do not influence the velocity field in our model.[6] The PDFs of x- and y-velocities resemble those of z-velocity as turbulence is isotropic. The $f_{S_e'}$ is narrower for Cases L and H compared to the forced cases (Fig. 18(b)) as fluctuations decay without forcing. We also see that the peaks are sharp for the $f_{S_e'}$ but rounded for the $f_{w'}$. Hence, there is greater likelihood of extreme values in the supersaturation field compared to the velocity field.

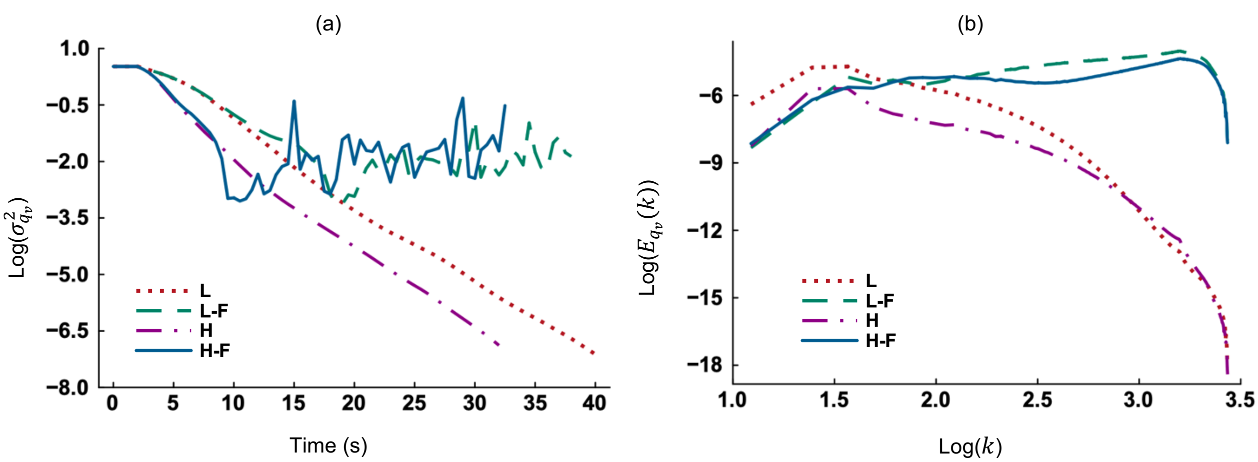


FIG. 17 (a) Time evolution of the variance of fluctuating vapor mixing ratio in Study II and (b) Spectra of fluctuating vapor mixing ratio field at 25 s in Study II. Scalar forcing is absent in Cases L and H. Cases L-F and H-F apply scalar forcing in the Fourier space with 'low' and 'high' velocity turbulence levels, respectively.

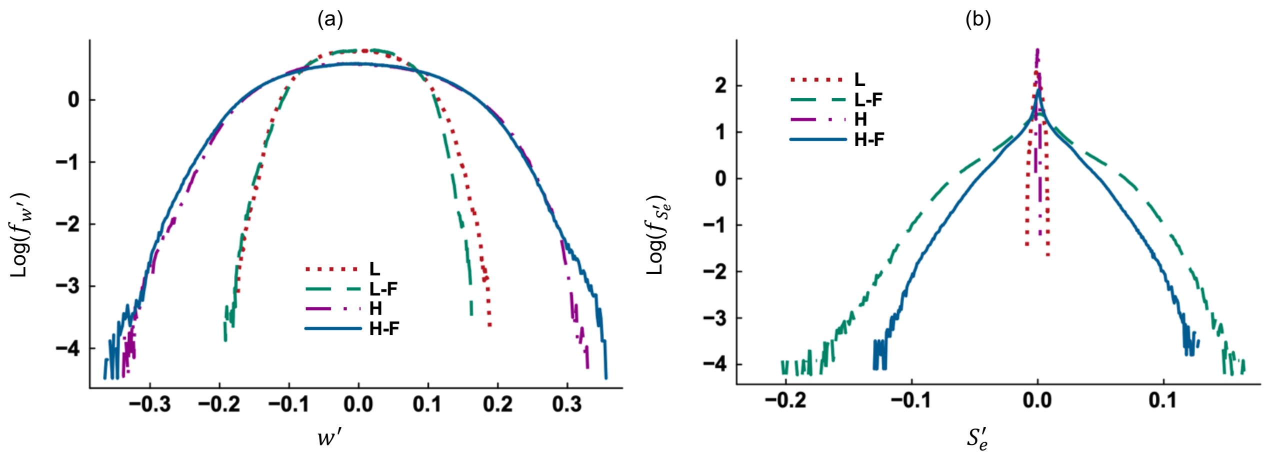


FIG. 18 Log PDFs of the (a) fluctuating z-velocity (w') and (b) fluctuating environmental supersaturation (S_e') at 25 s in Study II.

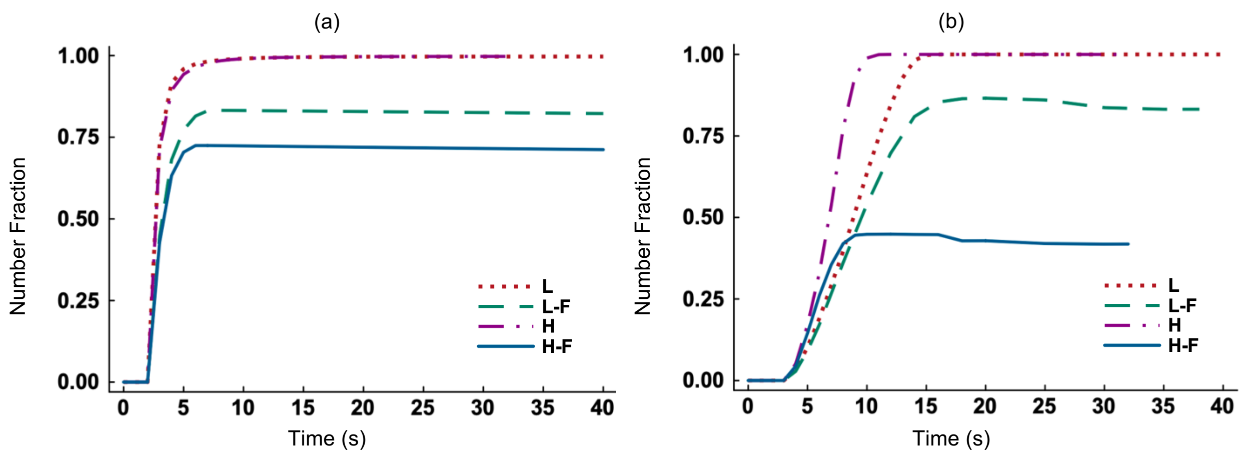


FIG. 19 Time evolution of the (a) number fraction of activated aerosol particles in Study I and (b) number fraction of deactivated cloud droplets in Study II. Scalar forcing is absent in Cases L and H. Cases L-F and H-F apply scalar forcing in the Fourier space with 'low' and 'high' velocity turbulence levels, respectively.

The intensity of flow turbulence as well as the magnitude of scalar forcing influence the microphysics of aerosols and cloud droplets. Figure 19(a) shows the number fractions of activated aerosol particles for the four cases. The activation fraction is slightly smaller for Case H compared to Case L between 3 s to 7 s. Higher turbulence leads to stronger mixing and causes most particles to grow evenly instead of some particles growing fast and crossing the activation barrier. Thus, activation fraction is reduced if the turbulence level is high and scalar forcing is absent (Case H). At long time, the activation fractions for Cases L and H are equal. We also see in Fig. 19(a) that fewer aerosol particles activate into cloud droplets for the forced cases (H-F and L-F) compared to the unforced cases (H and L). As scalar forcing leads the system towards equilibrium faster, it suppresses phase change (condensation and activation) in our model. The deactivation profiles in Fig. 19(b) show that deactivation is faster for Case H compared to Case L. Because of homogenization, more cloud droplets experience partial evaporation rather than some of them evaporating fully. This increases the number of cloud droplets that shrink below the critical radii and deactivate in Case H. Again, scalar forcing slows down phase change in Study II (evaporation) and reduces deactivation fraction. It is evident that both activation and deactivation fractions are smaller for Case H-F compared to Case L-F (Figs. 19(a)-(b)). This is because rapid mixing (high turbulence level) weakens phase change in our model.

V. CONCLUDING REMARKS

Using a three-dimensional finite difference based direct numerical simulation (DNS) model, we have examined the role of scalar forcing during condensation (activation) and evaporation (deactivation). The key findings from our simulations are itemized below.

- The mean and fluctuating scalar fields evolve from nonzero initial conditions. Forcing of the scalar fields leads scalar fluctuations to statistical stationarity (Figs. 3 and 7) and drives mean scalar fields to equilibrium faster (Figs. 5 and 10).
- While dissipation rates in the initial scalar fields are known, the mean scalar gradient β and the scalar variance α are not known a priori. Hence, it is easier to control the magnitude of forcing using the Fourier method than the other two methods. The magnitude of forcing is higher with mean gradient forcing compared to Fourier and linear forcings in our model.
- Unlike the mean velocity field, the mean scalar fields are not zero in our model. Hence, forcing increases the rate of change in the mean scalar fields. Note that rapid change in the mean temperature and vapor mixing ratio takes place during intense mixing between cloud and environment at the cloud edge.[47]
- The scalar fields, when coupled with particles, have reduced diffusivity and higher Scmidt number. Also, their spectral profiles include a viscous-convective regime (Figs. 4(a)-(b)) that falls between the inertial-convective and dissipation ranges. The spectra of unforced and forced scalar fields differ at the intermediate and small scales because forcing makes the viscous-convective regime dominant.
- No phase changes (condensation and evaporation) occur, and the mean vapor mixing ratio remains unchanged with time (Fig. 6(a)) in absence of particles. Scalar forcing does not cause deviation from homogeneity and isotropy in the scalar fields if particles are absent (Fig. 6(b)).
- The PDFs of fluctuating scalar fields have high kurtosis and are not Gaussian if forcing is applied (Fig. 9). Mean gradient forcing produces extreme fluctuations in our model leading to super-adiabatic droplet growth (Fig. 13(e)) which is often observed in stratocumulus clouds.[44]

- Scalar forcing reduces the time available for phase change. So, condensational growth is suppressed (Fig. 11(b)) and fewer aerosol particles activate into cloud droplets (Fig. 11(a)). Similarly, evaporation is suppressed (Fig. 14(b)) and fewer cloud droplets deactivate into aerosol particles (Fig. 14(a)).
- Activation and deactivation are mean-dominated at early times (Figs. 12(a) and 15(a)). At long time, activation and deactivation are fluctuation-dominated (Figs. 12(b) and 15(b)) if scalar forcing is present.
- Scalar forcing affects thermodynamic fluctuations at the small scales while momentum forcing does so at the large scales (Fig. 17(b)). Manton[48] found that both small-scale and large-scale fluctuations in the supersaturation broaden the radius PDFs of cloud droplets. This observation is consistent with our findings (Figs. 13 and 16).
- Higher velocity turbulence homogenizes scalar fluctuations if scalar forcing is absent. On the other hand, scalar forcing counteracts turbulent homogenization (Fig. 17(a)).
- By employing a DNS model with momentum and scalar forcings, we compare the effects of velocity turbulence and scalar fluctuations on microscale activation and deactivation (Figs. 19(a)-(b)). This is a novel contribution of the present study.
- The magnitude of thermodynamic fluctuations varies in natural clouds.[39] We show that a DNS model with scalar forcing can maintain thermodynamic fluctuations at desired levels and produce microphysical results that agree with physical observations.

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DATA AVAILABILITY

The data that support the findings of this work are available from the corresponding author upon reasonable request.

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